

1(a)	A trachea ; A windpipe I ref. to cartilage rings B bronchiole ; I respiratory / terminal , before bronchiole C alveoli ; A alveolus / air sac / alveolar sac	3
1(b)	<i>any one from</i> lung cancer ; emphysema ; <u>chronic bronchitis</u> ; COPD ; cystic fibrosis ; AVP ; e.g. asthma <i>must be non-infectious</i>	1
1(c)	<i>Plasmodium, falciparum / vivax / ovale / malariae ;</i>	1
1(d)	<i>any three from</i> <i>pathogen</i> (because) four / several, different, causative species / AW, (with different antigens) ; (pathogen has) many antigens as it is eukaryotic / many genes coding for antigens ; ref. to mutation changing antigens ; pathogen has different stages of life cycle (within human) with different antigens ; I ref. to antigenic shift antigenic concealment / pathogen spends part of life cycle within host cells / AW ; A short time in blood plasma <i>vaccine</i> ref. to more than one type of vaccine needed ; suggestion that antigens used have not given the strongest immune response ; <i>vaccination programme</i> ref. to only trial programmes ; A programmes not global ref. to not achieving herd immunity / not enough people (successfully) vaccinated ; detail ; e.g. lack of willingness to be vaccinated lack of trained people to give vaccines poor nutrition and poor immune response	3

Question	Answer	Marks
1(e)	<i>allow mosquitoes / vectors, for Anopheles</i> <i>pathogen / parasite, for Plasmodium</i> <i>any four from (max 1 for list of factors without explanation)</i> <i>assume in context of comparison with area Q</i> <i>ref. to a suitable climate for Anopheles, e.g. tropical / subtropical / hot and humid ;</i> <i>presence of Anopheles species which carry Plasmodium ;</i> <i>ref. Anopheles breeding sites present / (many) water sources (suitable for breeding) ;</i> <i>difficulty in controlling / lack of control, of Anopheles breeding sites ; AW, e.g. stagnant water not drained</i> <i>lack of / cannot afford, spraying with, insecticide / pesticide / repellents ;</i> <i>insecticide-resistant forms of Anopheles exist ;</i> <i>lack of / insufficient use of, mosquito nets ;</i> <i>mosquito nets not impregnated with insecticide ;</i> <i>lack of, artemisinin drugs / drugs to treat disease ;</i> <i>drug-resistant forms of Plasmodium exist ;</i> <i>idea of Plasmodium requiring temperatures of approximately 20°C (in vector) ;</i> <i>A area P has temperature / conditions for survival of Plasmodium</i> <i>ref. to difficulties in diagnosing disease (so delay / no treatment) ; e.g. lack of diagnostic tests lack of trained personnel</i> <i>AVP ;;</i> <i>e.g. some countries in area Q have Anopheles but control is good,</i> <i>ref. to feature of Anopheles species present in area P</i> <i>feed at all times during the day</i> <i>feed only on humans</i> <i>good host for reproduction of Plasmodium</i> <i>poor education so people don't know how to protect themselves from malaria</i>	4